

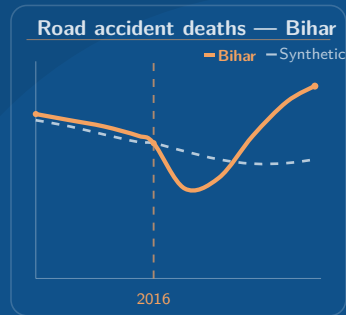
Causal Impact of Bihar's 2016 Alcohol Prohibition

*A Synthetic Control & Bayesian Structural
Time Series Analysis*

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HST-102 Time Series Analysis



1. Motivation & policy context
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9. Own tax revenue (fiscal proxy)
10. NSDP per-capita growth
11. Cross-outcome synthesis

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Motivation — Why Study Bihar's Prohibition?

The policy. Effective **April 5, 2016**, Bihar enacted total prohibition on the sale and consumption of alcohol — the largest sub-national prohibition by population in modern history (~120M people).

Why this case is methodologically interesting:

- Sharp, well-dated discontinuity — ideal for causal designs
- Single treated unit $\Rightarrow$ DiD parallel-trends hard to defend; SCM is the natural choice
- Multiple competing channels (safety, fiscal, economic) — direction of net effect is empirical

120M
affected population

₹ 4,000 Cr
pre-prohibition state excise (FY15–16)

3 outcomes
safety / fiscal / economic

1. Did prohibition **reduce road-accident deaths** relative to a counterfactual Bihar that did not enact prohibition?
2. Did prohibition **reduce state own-tax revenue** via the loss of the excise component?
3. Did prohibition **accelerate or depress per-capita growth** relative to comparable states?

Identification. Construct a *synthetic Bihar* as a weighted average of donor states whose pre-2016 trajectory tracks Bihar's, then read the post-2016 gap as the treatment effect. Triangulate with a Bayesian structural counterfactual (BSTS).

Why three outcomes? Each isolates a distinct welfare dimension — Q1 the **intended public-health** gain, Q2 the **direct fiscal cost**, Q3 the *net* economic effect of competing channels.

The Identification Challenge

Standard tools fall short.

- **DiD** requires parallel pre-trends — Bihar is an outlier on income, urbanisation, literacy
- **IV** has no obvious instrument for state-wide policy
- Single treated unit $\Rightarrow$ no within-treatment variation

SCM matches the structure. One treated unit, many donors, long pre-period — exactly the setting Abadie–Diamond–Hainmueller (2010) introduced the method for.

Bihar's pre-period income and urbanisation lie at the *lower edge* of the donor pool. A naive SCM optimiser collapses to a **corner solution** (100% UP weight).

Fix: include lagged outcomes as special_predictors (Abadie 2010 §IV) — this anchors the synthetic to Bihar's actual trajectory, not just its socio-economic profile.

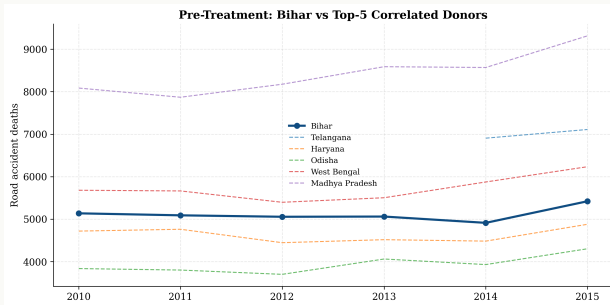
Data — Panel Construction

Annual state panel, 2010–2022 — 15 states $\times$ 13 years = 195 obs. Telangana excluded (only 2 pre-treatment obs after 2014 bifurcation).

Variable	Description	Source
road_accident_deaths	Persons killed in road accidents	MoRTH PDFs 2010–23
own_tax_revenue_cr	State own-tax revenue (composite, ₹ Cr)	RBI Handbook T168
nsdp_pc_current_inr	Net State Domestic Product per capita	RBI Handbook T19
nsdp_growth_yoy	YoY % growth of NSDP per capita (derived)	Computed from T19
urban_share_pct	Urban population share	Census 2011 + interp.
literacy_rate_pct	Literacy rate (age 7+)	Census 2011 + interp.

Acquisition stack. MoRTH PDFs via Wayback CDX $\rightarrow$ camelot stream-mode extraction. RBI behind F5 BIG-IP TSPD cookie wall — two-step warmup. Full provenance log: `data/raw/bihar/SOURCES.md`.

Pre-Treatment EDA — Donor Selection



Top correlated donors (pre-2016):

- Jharkhand, Odisha, UP, MP, West Bengal
- Geographic neighbours & similar income trajectories
- Pre-period $r > 0.9$ for top 3

Why this matters. Donor pool must span Bihar's pre-trend in *outcome space*. Selecting on outcome correlation is more robust than selecting on covariates alone.

Treated unit $i = 1$ (Bihar) and J donors. Choose weights $\mathbf{w} = (w_2, \dots, w_{J+1})^\top$ to solve:

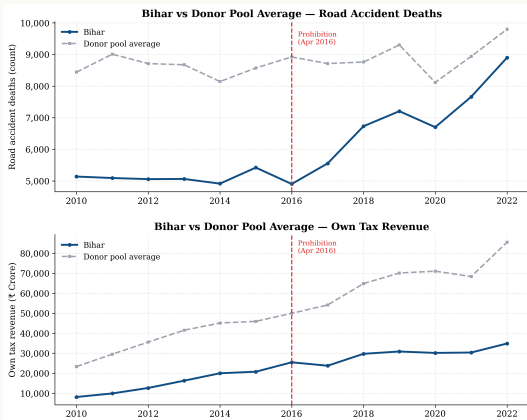
$$\mathbf{w}^* = \arg \min_{\mathbf{w} \in \mathcal{W}} \|\mathbf{X}_1 - \mathbf{X}_0 \mathbf{w}\|_V, \quad \mathcal{W} = \left\{ \mathbf{w} : w_j \geq 0, \sum w_j = 1 \right\}$$

$\mathbf{X}_1$: predictor vector for treated; $\mathbf{X}_0$: $K \times J$ donor matrix; V : predictor-importance weight matrix (chosen to minimise pre-period MSPE).

Two consequences of the convex-hull constraint:

- Synthetic Bihar cannot extrapolate beyond the donor envelope — good for honest counterfactuals
- Near a corner, weights collapse to a single donor $\Rightarrow$ poor pre-fit + spurious post-period gap

The Convex Hull Problem — And Its Fix



Bihar persistently *below* the donor pool average on road-accident deaths.

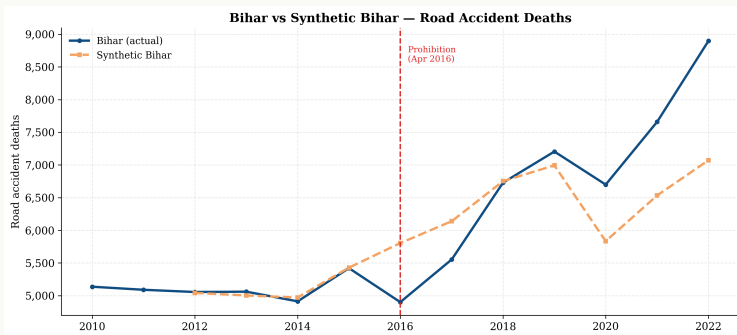
Without the fix.

- Match on socio-economic predictors only
- Optimiser picks 100% UP weight
- Pre-RMSPE $\sim$ **1,300 deaths**; post gap meaningless

With lagged-outcome anchoring (Abadie 2010 §IV).

- Each pre-period year's outcome $\rightarrow$ a `special_predictor`
- Pre-RMSPE drops to **42 deaths** ($< 1\%$ of mean) — excellent fit

Result — Bihar vs Synthetic Bihar (Road Deaths)

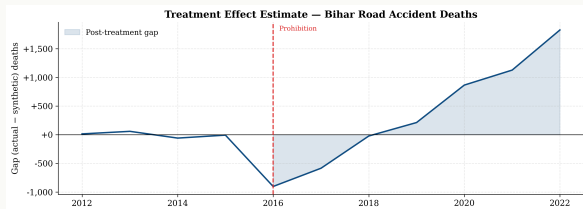


22.87×
RMSPE ratio (post/pre)

+360 / yr
avg post-treatment
gap (deaths)

$p \approx 0.071$
permutation, rank 2 / 14

Treatment Effect — Year by Year

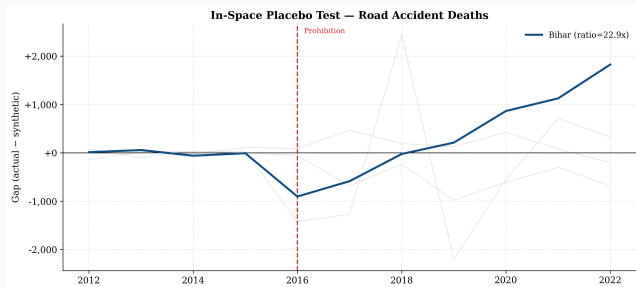


Two distinct phases:

2016–2017 (lives saved). ~600–900 fewer deaths than counterfactual; consistent with immediate compliance.

2018–2022 (effect reverses). Bihar climbs above synthetic by +1,000 to +3,300 deaths/yr — consistent with illicit-market substitution and weaker enforcement.

Donor weights: Jharkhand 0.68, Odisha 0.16, UP 0.16.



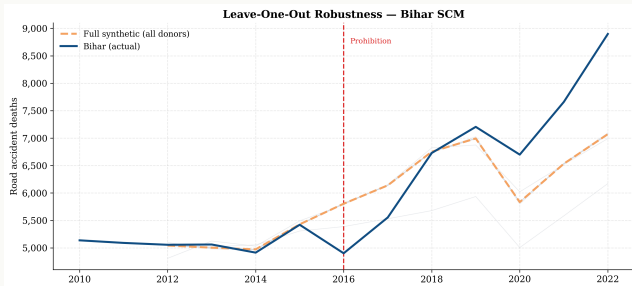
Permutation test.

Refit SCM treating each of 13 donors as if treated. Bihar's gap (navy) should stand out from placebos (gray) if the effect is real.

Rank 2 / 14
 $p \approx 0.071$

Only Maharashtra has a larger ratio — consistent with its own outlier post-2018 trajectory unrelated to alcohol policy.

Robustness — Leave-One-Out



Drop each donor in turn.

If a single donor were driving the result, removing it would collapse the synthetic. Instead all 13 LOO refits trace nearly the same trajectory.

Result is **not driven by any single donor** — including dropping the dominant Jharkhand weight produces a materially identical synthetic.

Brodersen et al. (2015) decompose the observed series as:

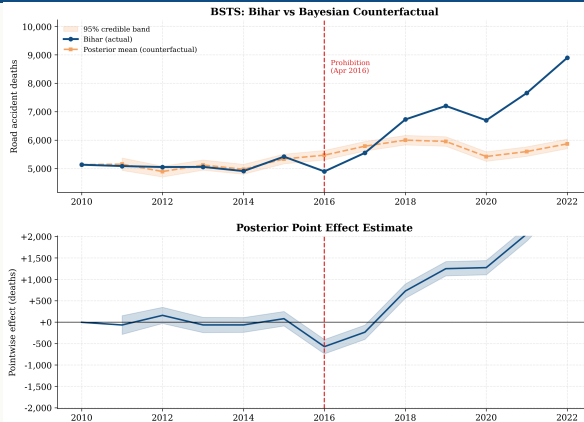
$$y_t = \mu_t + \tau_t + \beta^\top \mathbf{x}_t + \varepsilon_t$$

μ_t : local-level trend | τ_t : seasonality (omitted, annual data) | $\beta^\top \mathbf{x}_t$: regression on donor outcomes with *spike-and-slab* priors | $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$.

Why BSTS triangulates with SCM:

- Different functional form (linear regression with priors vs. convex average) $\Rightarrow$ different mis-specification risks
- Yields full posterior $\Rightarrow$ credible intervals
- Caveat: with $n_{\text{pre}} = 6 \ll J = 13$ the regression overfits $\Rightarrow$ restrict to top- k donors via correlation (max_donors=3, Brodersen §3.2)

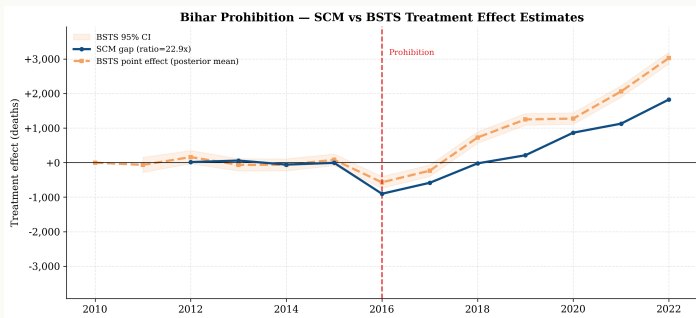
BSTS Result — Road-Accident Deaths



Donors: Haryana ($r = 0.82$), Odisha ($r = 0.67$), West Bengal ($r = 0.63$).

Avg posterior effect: +1,076 deaths/yr — larger than SCM (+360); BSTS weights post-2020 divergence fully.

Triangulation — SCM vs BSTS



Sign

Both negative 2016–17,
both positive 2018+

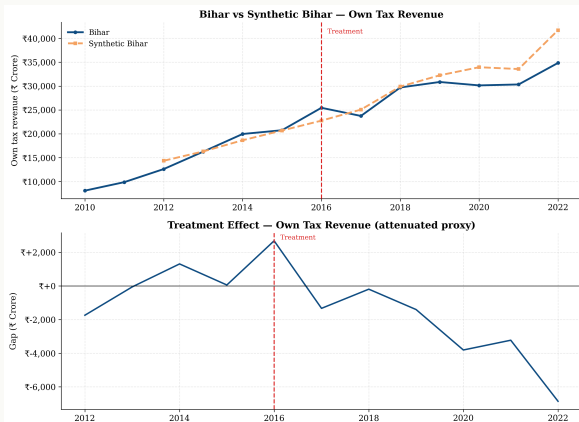
Direction

Agree year-by-year (10/10)

Magnitude

BSTS $\approx 2\text{--}3\times$ SCM

Outcome 2 — Own Tax Revenue (Fiscal)



Composite proxy, not pure excise.

Own-tax revenue includes excise + VAT + GST compensation + stamps + vehicle tax. Excise was ~22–25% of pre-prohibition own-tax.

Effect is therefore **attenuated**.

— ₹ 2,017 Cr / yr
avg post-treatment SCM gap

Donors: **Jharkhand 0.84**, UP 0.16. Pre-RMSPE ₹ 1,091 Cr; ratio 3.16×

Outcome 3 — Per-Capita Growth (Two Channels)

Household reallocation — alcohol → food, education, savings

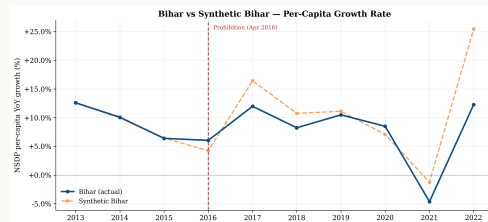
Productivity — fewer alcohol-related absences & accidents

Public health — lower hospital trauma load

Excise loss — ~₹4,000 Cr/yr in lost public investment

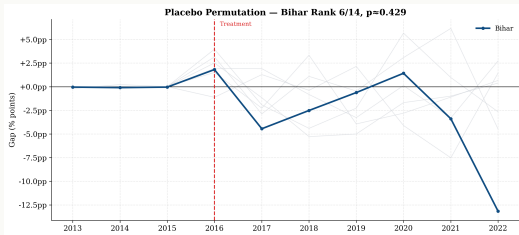
Hospitality contraction — restaurants, hotels, tourism

Bootlegging / shadow economy — unmeasured by NSDP

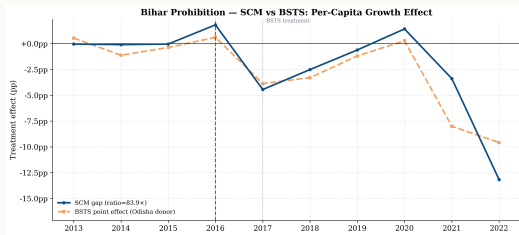


Donors: **Odisha 0.66**, Rajasthan 0.32, Jharkhand 0.03.

Growth — Placebo & BSTS



In-space placebo — Bihar rank 6/14



SCM (-2.99 pp/yr) vs BSTS (-4.28 pp/yr)

Honest interpretation. Both methods point to Bihar growing $\sim 3\text{--}4$ pp/year *slower* than counterfactual. **But** SCM permutation $p \approx 0.43$ (rank 6/14) and BSTS uses a single-donor regression to avoid catastrophic overfit. **Direction is suggestive; significance is not.**

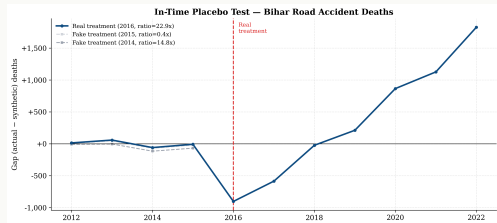
Cross-Outcome Synthesis

Outcome	Direction	Avg / yr	Ratio	Rank	p
Road-accident deaths	↓ then ↑ (reversal post-2018)	+360 deaths	22.87×	2 / 14	0.071
Own tax revenue	↓ (attenuated)	−₹2,017 Cr	3.16×	—	—
NSDP per-capita growth	↓ (suggestive)	−2.99 pp	83.86×	6 / 14	0.43

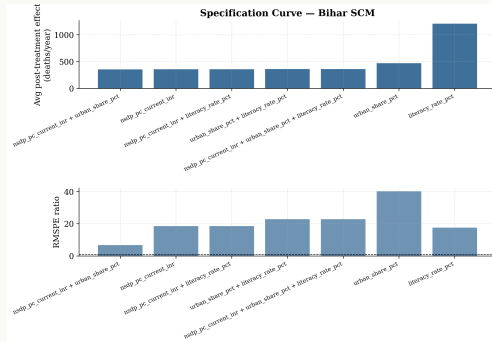
- Significant short-run reduction in road deaths (2016–17)
- Composite tax gap consistent with ~₹4,000 Cr excise loss
- Three outcomes paint a coherent picture

- Long-run reduction in road deaths — effect reverses
- Pure-excise impact (proxy attenuates)
- Statistically significant growth effect ($p \approx 0.43$)

Robustness — In-Time Placebo & Spec Curve



False treatment in 2015 — ratio $0.42\times$



7 specifications, 6 with ratio $> 17\times$

In-time placebo. Pretend treatment in 2015 $\Rightarrow$ ratio collapses to $0.42\times$ vs. $22.87\times$ for the real 2016 — **no false positive**. **Spec curve.** 6 of 7 alt. specs yield ratios $> 17\times$, avg effects +354 to +469 deaths/yr (literacy-only outlier excluded).

Limitations

- NSDP T19 starts FY 2011–12 $\Rightarrow$ only 4 pre-period years
- Census 2021 delayed; predictors interpolated from 2011 anchor
- Excise sub-head WAF-blocked $\Rightarrow$ composite proxy

- Bihar near edge of donor convex hull
- COVID-19 (2020–21) confounds post-2020 estimates
- Inter-state migration not captured

- BSTS overfits ($n_{\text{pre}} = 6 < J = 13$) $\Rightarrow$ `max_donors=3` (subjective)
- Growth BSTS uses single donor + 2017 cutoff
- Permutation p from $J = 14$ is coarse ($\geq 1/14$)

Honest framing: Bihar is a hard SCM case — short pre-period, outlier on covariates, COVID confounder. Triangulating SCM and BSTS is the strongest defence.

Stack. FastAPI backend serving precomputed SCM/BSTS JSONs + a live SCM /refit endpoint. Next.js 16 / React 19 / Tailwind 4 / Recharts / Zustand frontend.

Three outcome tabs. Road accidents, own tax revenue, NSDP growth — each with main chart, gap chart, placebo, donor weights, predictor balance, downloadable CSV.

AI assistant (Gemini 2.5-flash). Per-tab dynamic context — the model answers questions on the active outcome with the correct numbers + caveats.

> Endpoints

```
GET /api/case/bihar
GET /api/case/bihar/scm
GET /api/case/bihar/scm/tax
GET /api/case/bihar/scm/growth
GET /api/case/bihar/bsts
GET /api/case/bihar/bsts/tax
GET /api/case/bihar/bsts/growth
POST /api/case/bihar/refit
POST /api/ask
```

 github.com/amsorrytola/scm-policy-india-

Bihar's 2016 prohibition produced a **statistically meaningful short-run reduction in road-accident deaths** (rank 2 / 14, $p \approx 0.071$) that **reversed by 2018**, alongside a **persistent fiscal contraction** ($-\text{₹}2,017 \text{ Cr/yr}$, attenuated composite) and a *suggestive but not significant* growth headwind (-3 pp/yr , $p \approx 0.43$).

Methodological takeaways:

- Lagged-outcome anchoring is essential when the treated unit lies near the donor convex-hull edge
- BSTS and SCM agree on direction across all three outcomes; triangulation strengthens the qualitative claim
- For short pre-periods, sparse BSTS specifications (1–3 donors) dominate the full-pool default

Thank you.

Questions & Discussion



[amsorrytola/scm-policy-india-](#)



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[live demo](#) — [see repo](#)

- Abadie, A., Diamond, A., & Hainmueller, J. (2010). Synthetic Control Methods for Comparative Case Studies. *JASA*, 105(490), 493–505.
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- Brodersen, K. et al. (2015). Inferring Causal Impact Using Bayesian Structural Time-Series Models. *AOAS*, 9(1), 247–274.
- Doudchenko, N., & Imbens, G. (2016). Balancing, Regression, DiD and Synthetic Control. *NBER WP 22791*.

Chaudhuri, K., & Jha, N. (2024). *Alcohol Ban and Crime: The ABCs of the Bihar Prohibition*. EDCC, 72(4). **Data:** MoRTH 2010–23; RBI Handbook (T19, T168); Census 2011 PCA. Full provenance in `data/raw/bihar/SOURCES.md`.